

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset L^AT_EX solutions.

0.a

For this problem we are going to define the CSP as follows:

$$(X, D, C).$$

Being X , D and C the Variables, Domains and Constraints

For each button $j \in \{1, \dots, m\}$, that will react on the light bulb, we define a variable

$$X_j \in \{0, 1\}.$$

Meaning that each variable can have two values:

$$X_j = \begin{cases} 1 & \text{if button } j \text{ is pressed,} \\ 0 & \text{otherwise.} \end{cases}$$

Then, it is important to define the Domain as:

$$D(X_j) = \{0, 1\} \quad \forall j \in \{1, \dots, m\}.$$

According to the data given, there is one constraint for each light bulb.

For each bulb $i \in \{1, \dots, n\}$, define

$$S_i = \{j \in \{1, \dots, m\} \mid i \in T_j\},$$

the set of buttons that toggle bulb i .

Then the scope would be:

$$\text{scope}(C_i) = \{X_j \mid j \in S_i\}.$$

$$C_i : \sum_{j \in S_i} X_j \equiv 1 \pmod{2}$$

This constraint enforces that bulb i , which is initially OFF, is toggled an odd number of times and therefore ends ON.

Let's suppose bulb 1 is controlled by buttons 1, 3, and 4, so

$$T_1 = \{1, 3, 4\}.$$

Then

$$S_1 = \{1, 3, 4\},$$

and the corresponding constraint is

$$X_1 + X_3 + X_4 \equiv 1 \pmod{2}.$$

For example:

$$(X_1, X_3, X_4) = (1, 0, 0)$$

satisfies the constraint since $1 \equiv 1 \pmod{2}$, so bulb 1 ends ON, while

$$(X_1, X_3, X_4) = (1, 1, 0)$$

violates the constraint since $2 \equiv 0 \pmod{2}$, so bulb 1 would end OFF.

0.b

i) The variables are binary:

$$X_1, X_2, X_3 \in \{0, 1\}.$$

The constraints are:

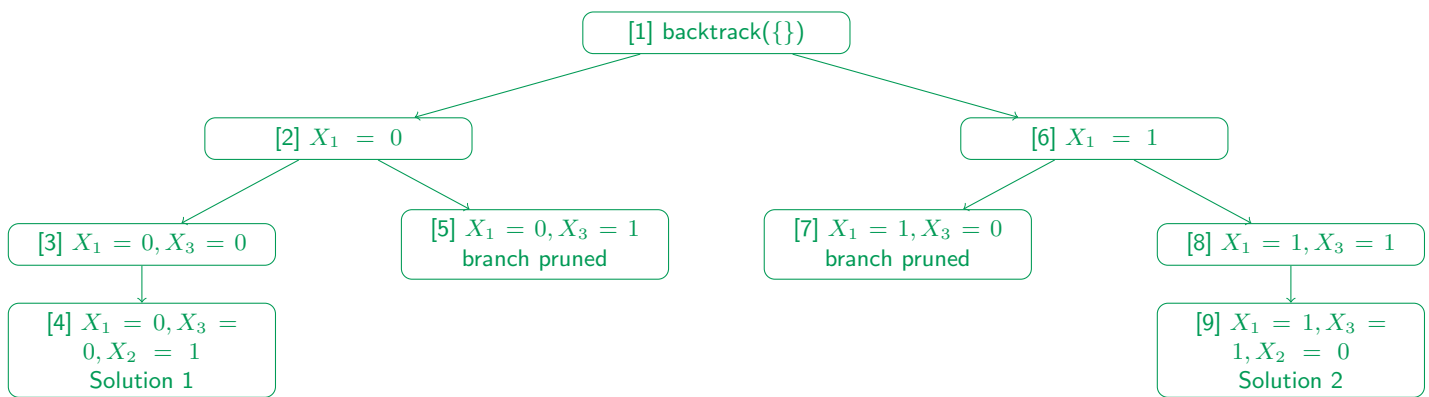
$$X_1 \oplus X_2 = 1 \quad \text{and} \quad X_2 \oplus X_3 = 1.$$

The first constraint implies $X_1 \neq X_2$, and the second implies $X_2 \neq X_3$. Therefore, once the value of X_2 is fixed, the values of X_1 and X_3 are uniquely determined.

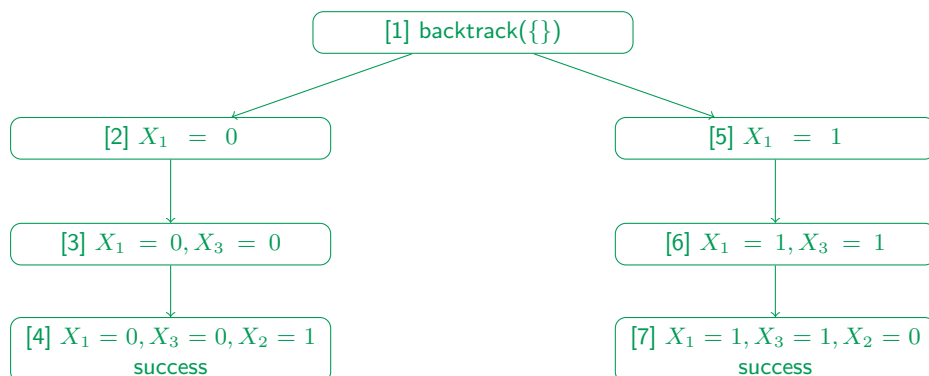
If $X_2 = 0$, then $X_1 = 1$ and $X_3 = 1$. If $X_2 = 1$, then $X_1 = 0$ and $X_3 = 0$, being the contrary to the first assignment, logically.

Both assignments satisfy all constraints, and there are no other possible assignments. So there would be 2 different assignments.

ii) Using the fixed ordering X_1, X_3, X_2 with XOR factors, the search explores all assignments to X_1 and X_3 first. When $X_1 \neq X_3$, no value of X_2 can satisfy both XOR constraints, so those branches end early (see backtrack [5] and [7]). Only the branches with $X_1 = X_3$ lead to a consistent X_2 , giving two solutions and a total of 9 backtrack calls.



iii) Using the fixed ordering X_1, X_3, X_2 with AC-3 lookahead, inconsistency can be detected earlier than in plain backtracking, reaching to a solution earlier. After assigning X_1 and X_3 , AC-3 enforces arc consistency on the remaining variable X_2 using the XOR constraints $X_1 \oplus X_2 = 1$ and $X_2 \oplus X_3 = 1$. If $X_1 \neq X_3$, these constraints impose incompatible requirements on X_2 , causing $\text{Domain}(X_2)$ to become empty and the branch to be pruned without a recursive call. If $X_1 = X_3$, AC-3 reduces $\text{Domain}(X_2)$ to a single value, leading to exactly one recursive call that produces a solution. Therefore, only 7 calls to Backtrack are made in total to enumerate all consistent assignments.



2.d

Profile (my_profile_test.txt)

```
# These would be the number of units that I would take
minUnits 6
maxUnits 9

# Here the quarters to enroll during 2018
# Note that this is an orientative example, I am not familiar how
# it works in Stanford/US

register Win2018
register Spr2018

# Winter 2018
request CS105 in Win2018
request CS149 in Win2018
request CS109L in Win2018

# Spring 2018
request CS101 in Spr2018
request CS178 in Spr2018
request CS181W in Spr2018
```

So the Best Schedule would be:

Quarter	Units	Course
Win2018	5	CS105
Win2018	1	CS109L
Spr2018	5	CS178
Spr2018	4	CS181W

3.a

I would change A by adding more residents. It's the only move that actually boosts capacity while keeping all five safety constraints untouched. A could be increased by reallocating or temporarily rotating residents from other centers, which doesn't necessarily require a big new budget. If possible, even a small or temporary staffing bump makes the CSP workable without relaxing any rules. If funding is tight, I would still aim for a modest increase in A or temporary coverage to close the feasibility gap while keeping all the rules as they are.